



**KERMOND CERTIFICATE OF EDUCATION
2023**

NAME:

SPECIALIST MATHEMATICS

Test 5 – CAS-Active

Theme: Differential Calculus

Reading time: 5 minutes

Writing time: 30 minutes

QUESTION AND ANSWER BOOK

Structure of book

<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
4	4	17

- Students are permitted to bring into the examination room: pens, pencils, highlighters, erasers, sharpeners, rulers, a protractor, set-squares, aids for curve sketching, one bound reference, one approved technology (calculator or software) and, if desired, one scientific calculator. Calculator memory DOES NOT need to be cleared. For approved computer-based CAS, full functionality may be used.
- Students are NOT permitted to bring into the examination room: blank sheets of paper and/or correction fluid/tape.

Materials supplied

- Question and answer book of 6 pages.

Instructions

- Write your **name** in the space provided above on this page.
- Write your **name** on the multiple-choice answer sheet
- Unless otherwise indicated, the diagrams are **not** drawn to scale.
- All written responses must be in English.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the assessment area.

The derivative of the function $f(x)$ is given by $f'(x) = \sin(x^3)$.

[illegible]

Consider the function $\sec(x)$ with restricted domain $\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$.

- [illegible]

- [illegible]

2 marks

[illegible]

[illegible]